2024 International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC) | 979-8-3503-7905-1/24/$31.00 ©2024 IEEE | DOI: 10.1109/ITC-CSCC62988.2024.10628188

A Quantum Circuit Design for Quantum Portfolio Optimization Problem

Vu Truc Quynh, Vu Tuan Hai, *Graduate Student Member, IEEE,* Le Vu Trung Duong, *Graduate Student Member, IEEE,* Pham Hoai Luan, *Member, IEEE,* and Yasuhiko Nakashima *Senior Member, IEEE*

***Abstract*—Portfolio optimization is an important problem in the quantum finance field, which aims to find optimal investment strategies that strike a balance between high returns and minimize risks with some constraints related. In the context of millions of assets, quantum methods such as Variational Quantum Eigensolver and Quantum Approximate Optimization Algorithm offer logarithm complexity compared with quadratic complexity in classical methods. However, those methods are designed for a massive range of applications and are not specified for any particular problem. Therefore, in this research, we present a quantum circuit design called QPO that has more functions for portfolio optimization, such as switch mode. We first tested the proposed circuit on 97 assets from 30 April 2023 to 30 June 2023. The results compared with classical solutions are consistent with an average Hamming distance 0.092.**

***Index Terms*—portfolio optimization, quantum finance, quantum circuit design**

I. Introduction

Nowadays, investors encounter with a increasingly wide range of choices. To find a profitable one, they have to consider those assets carefully. After that, they decide to buy the optimal one based on the result of the analysis and their risk appetite, this process is called portfolio optimization (PO). Modern portfolio theory (MPT) [1, 2] is one of the well-known mathematical theories being used to build models to enhance this essential process by quantifying the relationship of risk and return. This action also correlates to diversification, which spreads the risk over different entities and possibly creates higher profit [3]. The core idea of MPT is that higher risk does not always come with higher return. Take an example, there are many cases that investors have to suffer from not only higher risk but lesser return. MPT also points out that the choice of investors is dependent on their risk appetite, if there are two options with the same return, the investors will choose the ones having lower risk. MPT facilitates their choice by giving them an overview of risk-return combinations in different assets in their intended portfolio. It measures risk as variance of return and demonstrates the relationship of risk and return as different points on the plot.

In MPT, the expected return is estimated by mean, and the risk is defined as the standard deviation of the return list created by the combination of N assets' expected return, with N being the number of assets we analyze. For an asset i with open price op_i^t and close price cp_i^t for days $t \in [1, T]$, and the covariance σ_{ij} between assets i, j is the risk when enabling diversification. Expected return r_i^t, and covariance through T days are calculated as (1):

$$\begin{aligned} r_i^j &= \frac{cp_i^t - op_i^t}{op_i^t}; \\ \mu_i &= \frac{1}{T}\sum_{t=1}^{T} r_i^t; \sigma_{ij} = \frac{1}{T}\sqrt{\sum_{t=1}^{T}\left(r_i^t - \mu_i\right)\left(r_j^t - \mu_j\right)}. \end{aligned} \tag{1}$$

Then, denote that μ and σ are the mean expected return vector and covariance matrix of N assets. In this paper, we focus on dealing with the single-time-step process, which means the output of MPT is a list of recommended assets. Assume that assets $x := \{x_i\} \in X$, where x_i is the weight in allocating the resources that need to be picked to maximize the returns while minimizing the risk at step $T + 1$. The total cost (sum of asset's price $b_i \times$ by weighted parameter x_i) is constrained to be less than or equal to budget B. Our goal is minimizing the risk-return combination as an objective function in (2):

$$\begin{aligned} \mathcal{L}(x) &= q\mathrm{Var}[\sum_{i=1}^{N} r_i x_i] - \sum_{i=1}^{N} \mu_i x_i \\ &= q\sum_{i,j=1}^{N} x_i x_j \sigma_{ij} - \sum_{i=1}^{N} \mu_i x_i \\ &= q x^\intercal \sigma x - \mu^\intercal x \ (\text{s.t } b^\intercal x \leq B). \end{aligned} \tag{2}$$

where $q \in \mathbb{R}$ is known as the risk aversion parameter, it is normally chosen as 0.5. When investors can choose any volume for each asset to make up their portfolio, such that $x \in \mathbb{R}^N$. Then, with the analytical method, the solution of (2) is $x^* = (\sigma^{-1}\mu)/q$ $(\sigma \succeq 0)$ can be easily found via the derivative with respect to x. In the case of the discrete portfolio mentioned in [4, 5], x is now the binary string that lets us know whether to choose to own an asset or not, and the budget B is as simple as the number of invested assets. Therefore, the constraint optimization method can be applied.

II. Related works

A. Classical approach

A common way to solve a discrete PO problem is to view it as a convex optimization problem and use methods such as brute-force search (with time complexity $\mathcal{O}(2^N)$)

Authorized licensed use limited to: CBS Library. Downloaded on March 31,2026 at 21:06:24 UTC from IEEE Xplore. Restrictions apply.

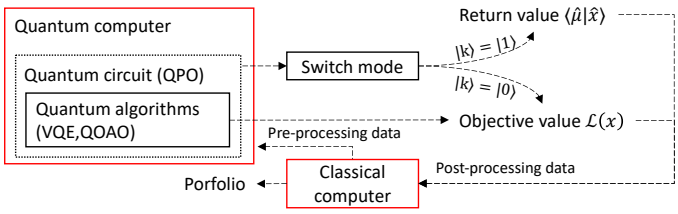

Fig. 1. Overview of portfolio optimization with hybrid quantum-classical's architecture.

linear programming [6], mean-variance optimization [7] (with time complexity $\mathcal{O}(N^3)$), simulated annealing [8], genetic algorithm [9], which is support by libraries, such as cvxpy [10, 11], and Scipy [12]. Overall, classical algorithms have polynomial time complexity with respect to a number of assets N, it's limited to the portfolio that has millions of assets because of the lack of classical computer resources. Therefore, quantum computing has been applied **to solve the discrete PO problem in millions of assets scenario with time complexity reduced to** $\mathcal{O}(\log N)$. Particularly, for processing N assets, the hybrid quantum-classical approach only needs $n = \log_2 N$ qubits, meaning the number of qubits only increases by one if the assets double.

B. Hybrid quantum-classical approach

Recently, the hybrid quantum-classical algorithm has been developed and applied widely, from quantum machine learning [13], optimization, chemistry [14], and material problems [15]. While quantum computers only conduct the high-complexity steps by quantum algorithms, classical computers are in charge of pre-processing and post-processing data, these two type of computers collaborate as described in Fig. 1. In detail, the classical computer calculates elements $\{\mu, \sigma\}$ and prepares states for quantum circuit; then, the quantum computer executes a circuit to yield measurement values that are relevant to return value or objective value with low time complexity. These values are sent to the classical computer again to turn into the final results.

1) Quadratic Unconstrained Binary Optimization (QUBO): To put $\mathcal{L}(x)$ into the quantum circuit, Abraham [16] has been re-written it as QUBO problem use the penalty term $P = \lambda(1^\intercal x - B)^2$ as (3):

$$\min_x \mathcal{L}'(x) = \min_x [\mathcal{L}(x) + P]. \tag{3}$$

The penalty coefficient, denoted as λ, is known as a Lagrange multiplier. It is the key parameter to state the problem as the QUBO. To solve QUBO, Abraham [16] must convert the variable x into a Pauli Z operator by transforming $x_i = \{0, 1\} \mapsto z_i = \{-1, 1\}$. After that, coefficients are re-arranged, and we obtain the Ising Hamiltonian. Minimizing Ising Hamiltonian that needs to minimize [17], as presented in Appendix A.

2) Quantum Approximate Optimization Algorithm (QAOA) [18, 19]: commonly is used algorithm to solve (3). First, we initialize qubits in a superposition state using Hadamard gates. Then, a series of quantum gates is applied to simulate the time evolution under the Hamiltonian $\hat{H}$ for a parameterized time and parameterized gates $U(\hat{H}, \beta, \gamma)$. Next, the quantum state is measured, and each measurement outcome represents a possible solution to the portfolio optimization problem. Finally, adjusting β, γ until getting the minimize expectation value of $\hat{H}$.

3) Variational Quantum Eigensolver (VQE): is a hybrid quantum-classical algorithm based on the variational principle can help us to formulate the PO problem by searching for the minimum energy eigenstate of some Hamiltonian [20, 21]. Specifically, given a Hamiltonian $\hat{H}$ representing the observable, and a parameterized wave function $|\hat{x}\rangle = U_{\hat{x}}(\boldsymbol{\theta})|0\rangle^{\otimes n}$ where $\langle .|.\rangle$ is the bra-ket notation [22], the ground energy E_0 is the minimized value as (4):

$$E_0 \equiv E_{\min} = \min_\theta \left\langle \mathbf{0} \left| U^\dagger_{\hat{x}}(\boldsymbol{\theta}) \hat{H} U_{\hat{x}}(\boldsymbol{\theta}) \right| \mathbf{0} \right\rangle, \tag{4}$$

where $U_{\hat{x}}(\boldsymbol{\theta})$ is a parameterized quantum circuit (PQC) with parameter $\boldsymbol{\theta}$. Look like QAOA, the circuit update $\boldsymbol{\theta}$ until getting E_0 via a zero-order gradient optimizer such as COBYLA and SLSQP [23], or first-order such as Adam [24] and QNG [25], as presented in Fig. 2. In detail, the Hamiltonian is mapped into the spin operator's bases. This base is formed by the tensor product of Pauli strings:

$$\hat{H} = \sum_j c_j P_j;\ P_j = \{I, X, Y, Z\}^{\otimes n}. \tag{5}$$

Then, rewrite (4) as:

$$\begin{aligned} E_0 &= \min_{\boldsymbol{\theta}} \left\langle \mathbf{0} \left| U^\dagger_{\hat{x}}(\boldsymbol{\theta}) \left(\sum_j c_j P_j \right) U_{\hat{x}}(\boldsymbol{\theta}) \right| \mathbf{0} \right\rangle \\ &= \min_{\boldsymbol{\theta}} \sum_j c_j \left\langle \mathbf{0} \left| U^\dagger_{\hat{x}}(\boldsymbol{\theta}) P_j U_{\hat{x}}(\boldsymbol{\theta}) \right| \mathbf{0} \right\rangle \\ &= \min_{\boldsymbol{\theta}} \sum_j c_j f_{P_j}(\boldsymbol{\theta}). \end{aligned} \tag{6}$$

The term f_{P_j} returns from the quantum simulator via multiple shots. Then, E_0 is computed by a summation on a classical computer. After achieving $\boldsymbol{\theta}^*$, $|\psi(\boldsymbol{\theta}^*)\rangle = U_{\hat{x}}|\mathbf{0}\rangle$ thus is the optimal portfolio.

III. Approach

The two mentioned methods, VQE and QAOA, find the optimal portfolio x^* and its objective value $\mathcal{L}'(x^*)$. However, when the user wants to know the return value or risk value, then they need to re-calculate the term $\mu^\intercal x$ or $x^\intercal \sigma x$ in (2), that consume at least $\mathcal{O}(N)$ time complexity at the classical computer. Therefore, we propose a quantum circuit design called QPO circuit (circuit for quantum portfolio optimization) that has more function, controlled by the control qubit k.

Authorized licensed use limited to: CBS Library. Downloaded on March 31,2026 at 21:06:24 UTC from IEEE Xplore. Restrictions apply.

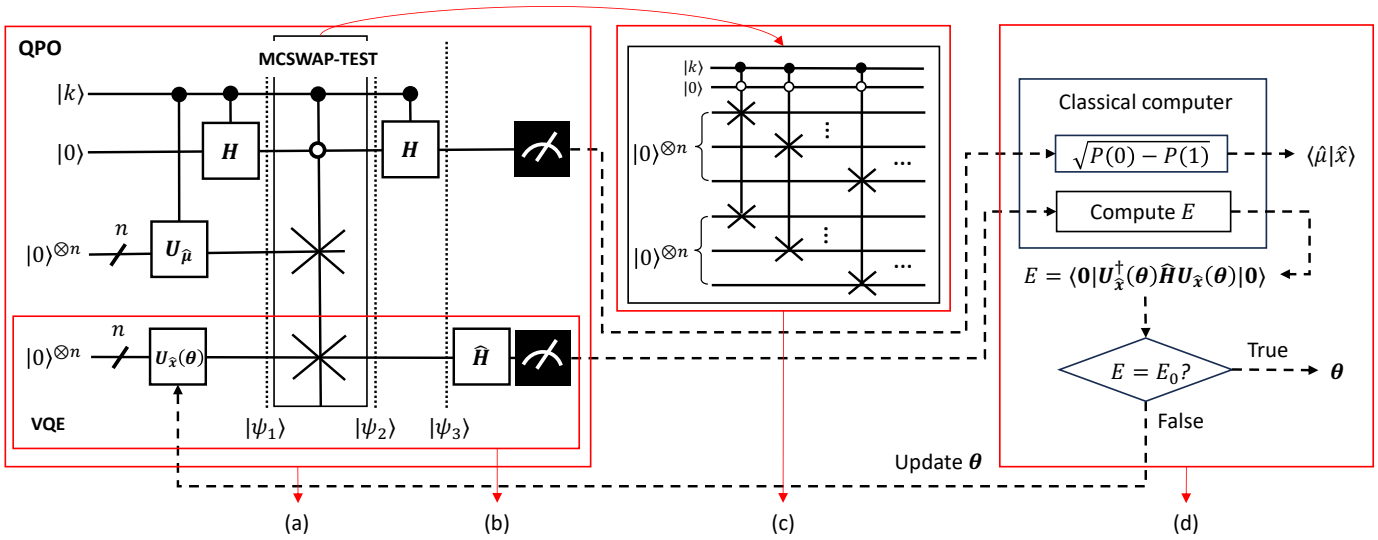

Fig. 2. (a) QPO design. The quantum state is initialized at $|k\rangle \otimes |0\rangle^{\otimes(2n+1)}$, then $\hat{x}$ and $\hat{\mu}$ are being prepared by $\langle qo|op\rangle$, the Hamiltonian $\hat{h}$ is also prepared at the last n qubits. The measurement device is put at the second qubits or last n qubits, which decides the type of output we want to take. If the first qubit is measured, the circuit outputs term $\langle\hat{x}|\sigma|\hat{x}\rangle$ and otherwise. (b) The sub-circuit in last n qubits is VQE circuit, during VQE process, $\boldsymbol{\theta}$ is updated iteratively until achieving E_0. (c) MCSWAP-TEST sub-circuit detail structure. (d) Post-processing by classical computer.

Depending on k, QPO operates as VQE/QAOA or calculates return value $\mu^{\intercal}x$. With N as the number of assets and $n = \log_2(N)$, it consumes n qubits to encode $\hat{x}$ and n qubits to encode $\hat{\mu}$, then the total number of qubits is $2n+2$ with an ancillary qubit for Multi-control SWAP-TEST (MCSWAP) and control qubit $|k\rangle$. We denote that $x = |x|\hat{x}$ and $\mu = |\mu|\hat{\mu}$ which mean $\hat{x}$ and $\hat{\mu}$ are normalized vectors. Then, return value is written as $(|x||\mu|)\langle\hat{\mu}|\hat{x}\rangle$. If $|k\rangle = |0\rangle$, QPO acts as n-qubits VQE and the MCSWAP-TEST sub-circuit has been disabled, otherwise, MCSWAP-TEST sub-circuit is active.

QPO circuit can be viewed as an extended version of VQE/QAOA, which includes the encoding circuits, MCSWAP-TEST sub-circuit, and Hamiltonian part. The encoding part is in charge of preparing quantum states $|\hat{x}\rangle = U_{\hat{x}}(\boldsymbol{\theta})|0\rangle^{\otimes n}$ and $|\hat{\mu}\rangle = U_{\hat{\mu}}|0\rangle^{\otimes n}$ by encoders $U_{\hat{x}}(\boldsymbol{\theta})$ and $U_{\hat{\mu}}$, respectively. While $U_{\hat{\mu}}$ is fixed from start, $U_{\hat{x}}(\boldsymbol{\theta})$ can be updated by $\boldsymbol{\theta}$ for VQE circuit. This task can be solved by various preparing methods, from basis encoding [26], divide-and-conquer encoding [27], amplitude encoding [28] to quantum compiling encoding [29]. Because we only use n qubits for each N-dimensional state, then amplitude encoding and quantum compiling are used. While quantum compiling offers polynomial-scale circuit depth but consumes initial resources for computing its parameters, amplitude offers exponential-scale circuit depth, which is not suitable for a large number of qubits. These two encoding methods are applied by using the existing package named $\langle qo|op\rangle$ [30] depending on how large the number of assets is. This package can also simulate the Hamiltonian part efficiently [31].

MCSWAP-TEST sub-circuit in Fig. 2 include n MCSWAP gates which act on wires $\{(0,1,2,2+n),(0,1,3,3+n),\cdots,(0,1,1+n,1+2n)\}$. This circuit swap positions between $|\hat{x}\rangle$ and $|\hat{\mu}\rangle$ when the first control qubit ($|k\rangle$) and the second control qubit are $|1\rangle$ and $|0\rangle$, respectively. MCSWAP-TEST sends measurement values $\{P(0), P(1)\}$ to classical computer, then classical computer computes the return value as $|x||\mu|\sqrt{P(0)-P(1)}$, which is detailed in Appendix. B.

IV. Experiments

The proposed circuit was simulated on a classical computer, Windows 11, with 13th Intel core i7 13700 (24 cores) and 32GB DDR4 RAM. The versions of Python and Qiskit are **3.10.9** and **0.45.1**, respectively. All codes and experimental results are available at https://github.com/tulliavu/QPO.

A. Dataset

This research use cryptocurrency price data to compare existing and new methods. The daily data are collected from Binance https://binance.com using klines, an open-source tool that uses Binance's publicly available APIs. According to its creator, this tool is intended to analyze past market behavior when providing us with information related to price, volume, duration of investigated interval. To explore the efficiency of the proposed approach, small-sized examples are considered by extracting 97 distinctive different assets each day in two months, from 30 April 2023 to 30 June 2023. USDT is the root since these coin related to this root are those stablecoins. In addition, these values are approximately equal to those of a US dollar in reality. Each of the asset is encoded based on their original source. There are open price at open time and close price at close time. During the day, there is also a record of highest price and lowest price. For each day t in this range $(0 \le t \le T)$, the performance of an asset is well represented by the fraction between the difference of closing and opening price within a day with its open price. Then, we take average

Authorized licensed use limited to: CBS Library. Downloaded on March 31,2026 at 21:06:24 UTC from IEEE Xplore. Restrictions apply.

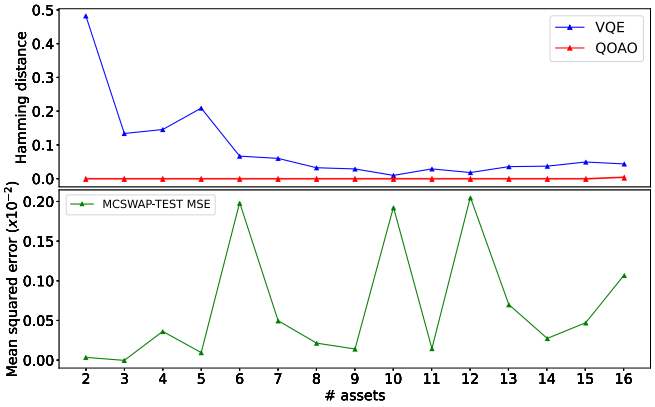

Fig. 3. (Top) Hamming distance between VQE/QAOA portfolio and target portfolio for each number of assets. (Bottom) MSE of MCSWAP-TEST sub-circuit.

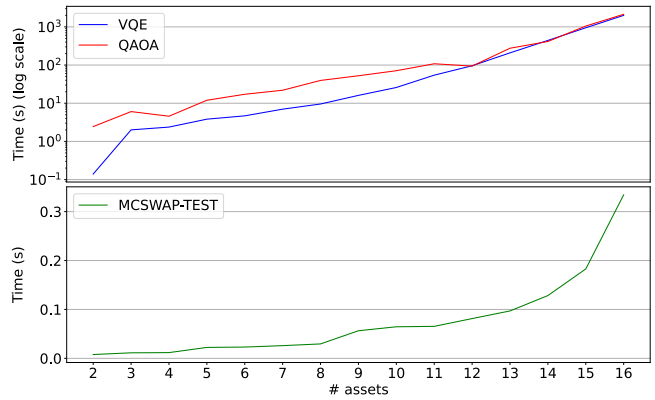

Fig. 4. Execution time of VQE, QAOA, and MCSWAP-TEST sub-circuit.

value of return in two months, as a result, each asset will have their own mean value.

B. Result

Denote that $x^*_{X_k}$ is the target portfolio over sub-asset $X_k (X_k \in X)$ from the linear programming method [6], where k is the number of assets, and X_k is chosen from our dataset randomly $\mathcal{N}$ times. Then, average Hamming distance [32] target portfolio $x^*_{X_k}$ with predicted portfolio x_{X_k} is computed as (7):

$$\hat{\mathcal{H}}_k = \frac{1}{\mathcal{N}} \sum_{X_k \subset X} \frac{\mathcal{H}(x_{X_k}, x^*_{X_k})}{|x_{X_k}|} \tag{7}$$

Our algorithms is evaluated from 2 to 16 assets with $\mathcal{N} = 100$ and budget $B = \lfloor |x_{X_k}|/2 \rfloor$. Fig. 3 shows the Hamming distance between QAOA or VQE portfolio and target portfolio, QAOA [18, 19] achieves approximate zero for all number of assets while VQE decreases its Hamming distance according to the number of assets. Note that the QPO circuit includes VQE and MCSWAP-TEST sub-circuit, then QPO is evaluated through VQE's Hamming distance and MCSWAP-TEST's mean square error (MSE). MSE is calculated between the exact and estimated value of term $\langle \hat{\mu} | \hat{x} \rangle$ which is under the acceptance threshold (10^{-2}). The simulation time in quantum circuits increases exponentially by the number of used qubits. Fig. 4 shows that in the case of a small number of assets, VQE in the QPO circuit runs faster than QAOA. Furthermore, MCSWAP-TEST does not consume computational time as much as VQE/QAOA.

V. Conclusion

In this paper, we proposed a quantum circuit design for portfolio optimization problem; this circuit can both conduct the optimization as VQE and calculate the return value, which has more function than normal VQE. The experiment on the real dataset showed the reliability of the proposed approach.

When scaling to larger portfolios, calculating the optimal portfolio in a classical computer might be much more expensive than using QPO, polynomial complexity compared with logarithm complexity. Still, as seen, even when quantum computing is used to solve those large combinatorial problems, it comes at the cost of needing complete certainty of the outcome. It is important to understand that QPO implementation requires the development of the encoder and Hamiltonian simulation, to reduce the quantum computation resources. In many cases, selecting an encoder will come from the experience of fitting against different types of portfolios and stock combinations. However, not all assets show similar behaviors. This will require exploring the vast potential encoder and Hamiltonian simulation technique and conducting experiments on larger assets.

Acknowledgment

This work was supported by JST-ALCA-Next Program Grant Number JPMJAN23F5, Japan. The research has been partly executed in response to the support of JSPS, KAKENHI Grant No. 22H00515, Japan.

References

[1] Harry Markowitz. "Portfolio Selection". In: *The Journal of Finance* 7.1 (1952), pp. 77–91. ISSN: 00221082, 15406261. URL: http://www.jstor.org/stable/2975974 (visited on 03/14/2024).

[2] Jon Lukomnik and James P Hawley. *Moving beyond modern portfolio theory: Investing that matters*. Routledge, 2021.

[3] M. Barkhagen et al. "Optimising portfolio diversification and dimensionality". In: *Journal of Global Optimization* 85.1 (Jan. 2023), pp. 185–234. ISSN: 1573-2916. DOI: 10.1007/s10898-022-01202-7. URL: https://doi.org/10.1007/s10898-022-01202-7.

Authorized licensed use limited to: CBS Library. Downloaded on March 31,2026 at 21:06:24 UTC from IEEE Xplore. Restrictions apply.

[4] Abhishek Gunjan and Siddhartha Bhattacharyya. "A brief review of portfolio optimization techniques". In: *Artificial Intelligence Review* 56.5 (May 2023), pp. 3847–3886. ISSN: 1573-7462. DOI: 10.1007/s10462-022-10273-7. URL: https://doi.org/10.1007/s10462-022-10273-7.

[5] Majid Zanjirdar. "Overview of Portfolio Optimization Models". In: *Advances in Mathematical Finance and Applications* 5.5 (1399). ISSN: 2538-5569. DOI: 10.22034/amfa.2020.674941. eprint: sanad.iau.ir/fa/Article/Download/915099. URL: sanad.iau.ir/fa/Article/915099.

[6] David Applegate et al. "Faster first-order primal-dual methods for linear programming using restarts and sharpness". In: *Mathematical Programming* 201.1 (Sept. 2023), pp. 133–184. ISSN: 1436-4646. DOI: 10.1007/s10107-022-01901-9. URL: https://doi.org/10.1007/s10107-022-01901-9.

[7] Jang Ho Kim et al. "Mean–variance optimization for asset allocation". In: *The Journal of Portfolio Management* 47.5 (2021), pp. 24–40.

[8] Thomas Guilmeau, Emilie Chouzenoux, and Victor Elvira. "Simulated Annealing: a Review and a New Scheme". In: *2021 IEEE Statistical Signal Processing Workshop (SSP)*. 2021, pp. 101–105.

[9] Kyle Erwin and Andries Engelbrecht. "Meta-heuristics for portfolio optimization". In: *Soft Computing* 27.24 (Dec. 2023), pp. 19045–19073. ISSN: 1433-7479. DOI: 10.1007/s00500-023-08177-x. URL: https://doi.org/10.1007/s00500-023-08177-x.

[10] Steven Diamond and Stephen Boyd. "CVXPY: A Python-embedded modeling language for convex optimization". In: *Journal of Machine Learning Research* 17.83 (2016), pp. 1–5.

[11] Akshay Agrawal et al. "A rewriting system for convex optimization problems". In: *Journal of Control and Decision* 5.1 (2018), pp. 42–60.

[12] Pauli Virtanen et al. "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python". In: *Nature Methods* 17 (2020), pp. 261–272. DOI: 10.1038/s41592-019-0686-2.

[13] Maria Schuld and Francesco Petruccione. *Machine learning with quantum computers*. Springer, 2021.

[14] Seunghoon Lee et al. "Evaluating the evidence for exponential quantum advantage in ground-state quantum chemistry". In: *Nature Communications* 14.1 (Apr. 2023), p. 1952. ISSN: 2041-1723. DOI: 10.1038/s41467-023-37587-6. URL: https://doi.org/10.1038/s41467-023-37587-6.

[15] Nhan Tri Tran, Hoang Thanh Nguyen, and Lan Nguyen Tran. "Reaching High Accuracy for Energetic Properties at Second-Order Perturbation Cost by Merging Self-Consistency and Spin-Opposite Scaling". In: *The Journal of Physical Chemistry A* 128.8 (Feb. 2024), pp. 1543–1549. ISSN: 1089-5639. DOI: 10.1021/acs.jpca.3c07450. URL: https://doi.org/10.1021/acs.jpca.3c07450.

[16] Abraham P Punnen. "The quadratic unconstrained binary optimization problem". In: *Springer International Publishing* 10 (2022), pp. 978–3.

[17] Frank Phillipson and Harshil Singh Bhatia. "Portfolio Optimisation Using the D-Wave Quantum Annealer". In: *Computational Science – ICCS 2021*. Ed. by Maciej Paszynski et al. Cham: Springer International Publishing, 2021, pp. 45–59.

[18] Kostas Blekos et al. "A review on quantum approximate optimization algorithm and its variants". In: *arXiv preprint arXiv:2306.09198* (2023).

[19] Gopal Chandra Santra et al. "Squeezing and quantum approximate optimization". In: *Phys. Rev. A* 109 (1 Jan. 2024), p. 012413. DOI: 10.1103/PhysRevA.109.012413. URL: https://link.aps.org/doi/10.1103/PhysRevA.109.012413.

[20] Huanfeng Zhao, Peng Zhang, and Tzu-Chieh Wei. "A universal variational quantum eigensolver for non-Hermitian systems". In: *Scientific Reports* 13.1 (Dec. 2023), p. 22313. ISSN: 2045-2322. DOI: 10.1038/s41598-023-49662-5. URL: https://doi.org/10.1038/s41598-023-49662-5.

[21] Shengbin Wang et al. "Variational quantum eigensolver with linear depth problem-inspired ansatz for solving portfolio optimization in finance". In: *arXiv preprint arXiv:2403.04296* (2024).

[22] Jun John Sakurai and Jim Napolitano. *Modern quantum mechanics*. Cambridge University Press, 2020.

[23] Xavier Bonet-Monroig et al. "Performance comparison of optimization methods on variational quantum algorithms". In: *Phys. Rev. A* 107 (3 Mar. 2023), p. 032407. DOI: 10.1103/PhysRevA.107.032407. URL: https://link.aps.org/doi/10.1103/PhysRevA.107.032407.

[24] Mohamed Reyad, Amany M. Sarhan, and M. Arafa. "A modified Adam algorithm for deep neural network optimization". In: *Neural Computing and Applications* 35.23 (Aug. 2023), pp. 17095–17112. ISSN: 1433-3058. DOI: 10.1007/s00521-023-08568-z. URL: https://doi.org/10.1007/s00521-023-08568-z.

[25] James Stokes et al. "Quantum Natural Gradient". In: *Quantum* 4 (May 2020), p. 269. ISSN: 2521-327X. DOI: 10.22331/q-2020-05-25-269. URL: https://doi.org/10.22331/q-2020-05-25-269.

[26] Seth Lloyd et al. "Quantum embeddings for machine learning". In: *arXiv preprint arXiv:2001.03622* (2020).

[27] Israel F. Araujo et al. "A divide-and-conquer algorithm for quantum state preparation". In: *Scientific Reports* 11.1 (Mar. 2021), p. 6329. ISSN: 2045-2322. DOI: 10.1038/s41598-021-85474-1. URL: https://doi.org/10.1038/s41598-021-85474-1.

[28] Maria Schuld and Francesco Petruccione. *Supervised learning with quantum computers*. Vol. 17. Springer, 2018.

Authorized licensed use limited to: CBS Library. Downloaded on March 31,2026 at 21:06:24 UTC from IEEE Xplore. Restrictions apply.

[29] Vu Tuan Hai, Nguyen Tan Viet, and Le Bin Ho. "Variational preparation of entangled states on quantum computers". In: *arXiv preprint arXiv:2306.17422* (2023).

[30] Vu Tuan Hai, Nguyen Tan Viet, and Le Bin Ho. "⟨qo—op⟩: A quantum object optimizer". In: *SoftwareX* 26 (2024), p. 101726. ISSN: 2352-7110. DOI: https://doi.org/10.1016/j.softx.2024.101726. URL: https://www.sciencedirect.com/science/article/pii/S2352711024000979.

[31] Laura Clinton, Johannes Bausch, and Toby Cubitt. "Hamiltonian simulation algorithms for near-term quantum hardware". In: *Nature Communications* 12.1 (Aug. 2021), p. 4989. ISSN: 2041-1723. DOI: 10.1038/s41467-021-25196-0. URL: https://doi.org/10.1038/s41467-021-25196-0.

[32] Francesco Pappalardo et al. "HAMFAST: Fast Hamming Distance Computation". In: *2009 WRI World Congress on Computer Science and Information Engineering*. Vol. 1. 2009, pp. 569–572. DOI: 10.1109/CSIE.2009.223.

APPENDIX

A. Hamiltonian for covariance matrix

In particular, a quite general formulation of the constrained quadratic problem is considered, which is translated into QUBO by the binary encoding of variables and by including constraints in the objective function. This is converted into a set of quantum operators (Ising Hamiltonian), whose minimum eigenvalue is found by VQE and corresponds to the optimal solution.

$$\min_{x_i\in\{0,1\}} \mathcal{L}'(x) \mapsto \min_{z_i\in\{1,-1\}} \sum_{i,j} J_{ij} z_i z_j + \sum_i h_i z_i + C,$$

with $J_{i,j}$ is the coupling term between two spin variables.

The calculation process for converting $x \in \{0,1\}$ to $z \in \{1,-1\}$, note that 1 and -1 is the eigenvalue for eigenstate $|0\rangle$ and $|1\rangle$ respectively, 0 is converted to 1 and 1 is converted to -1, rather than 0 to -1 and 1 to 1. The term $x^\intercal \sigma x$ can be written as:

$$\begin{aligned} x^\intercal \sigma x &= \sum_{i=1}\sum_{j=1,j>i} \sigma_{ij} x_i x_j \\ &+ \sum_{i=1}\sum_{j=1,j<i} \sigma_{ij} x_i x_j + \sum_{i=1} \sigma_{ii} x_i^2 \\ &= 2\sum_{i=1}\sum_{j=1,j>i} \sigma_{ij} x_i x_j + \sum_{i=1} \sigma_{ii} x_i^2 \quad (\sigma_{ij} = \sigma_{ji}). \end{aligned}$$

Now $x_i = (1 - z_i)/2$, $z_i^2 = 1$ and $\sum_i \sum_{j=1,j>i} \sigma_{ij} (-z_i - z_j) = -\sum_i \sum_{j=1,j\neq i} \sigma_{ij} z_i$, then:

$$\begin{aligned} x^T \sigma x = \frac{1}{2}[&\sum_i (\sum_{j=1,j>i} \sigma_{ij}) z_i z_j \\ &- \sum_{i=1} (\sigma_{ii} + \sum_{j=1,j\neq i} \sigma_{ij}) z_i \\ &+ (\sum_{j=1,j>i} \sigma_{ij} + \sum_{i=1} \sigma_{ii})]. \end{aligned}$$

Then. the Ising Hamiltonian has all its coefficient. Note that to minimize this Hamiltonian, we can drop the constant and only consider the quadratic $z_i z_j$ and linear z_i terms.

B. SWAP-test procedure

Initially, we apply the amplitude encoding method for preparing $|\hat{x}\rangle$ and quantum compiling for preparing $|\hat{\mu}\rangle$. Because $\hat{\mu}$ is fixed during the optimization process, quantum compiling is given priority with low depth despite the fact that it has resources for initialization. After that, the MCSWAP-TEST sub-circuit includes two control Hadamard gates and n multi-control SWAP gates.

$$\begin{aligned} |\psi_1\rangle &= |1\rangle \otimes (H|0\rangle) \otimes (U_{\hat{x}}|\mathbf{0}) \otimes (U_{\hat{\mu}}|\mathbf{0}\rangle) \\ &= \frac{1}{\sqrt{2}} |1\rangle \otimes (|0\rangle|\hat{x}\rangle|\hat{\mu}\rangle + |1\rangle|\hat{x}\rangle|\hat{\mu}\rangle). \end{aligned}$$

Whereas the action of n MCSWAP gates makes:

$$\begin{aligned} |\psi_2\rangle &= \text{MCSWAP}|\psi_1\rangle \\ &= \frac{1}{\sqrt{2}} |1\rangle \otimes (|0\rangle|\hat{x}\rangle|\hat{\mu}\rangle + |1\rangle|\hat{\mu}\rangle|\hat{x}\rangle). \end{aligned}$$

The last control Hadamard gate act on the second qubit again makes the system into:

$$\begin{aligned} |\psi_3\rangle = \frac{1}{2}|1\rangle \otimes ((&|0\rangle|\hat{x}\rangle|\hat{\mu}\rangle + |1\rangle|\hat{x}\rangle|\hat{\mu}\rangle) \\ &+ (|0\rangle|\hat{\mu}\rangle|\hat{x}\rangle - |1\rangle|\hat{\mu}\rangle|\hat{x}\rangle)). \end{aligned}$$

When we measure the first two qubits, there are two possible states that can receive:

$$\begin{aligned} P(q_0 q_1 = 10) &= \frac{1}{2} + \frac{1}{2}|\langle\hat{x}|\hat{\mu}\rangle|^2, \\ P(q_0 q_1 = 11) &= \frac{1}{2} - \frac{1}{2}|\langle\hat{x}|\hat{\mu}\rangle|^2. \end{aligned}$$

Since $x, \mu \in \mathbb{R}_+^N$, the absolute notation can be removed:

$$\begin{aligned} \langle\hat{x}|\hat{\mu}\rangle &= \sqrt{P(q_0 q_1 = 10) - P(q_0 q_1 = 11)} \\ &= \sqrt{P(0) - P(1)}. \end{aligned}$$

Authorized licensed use limited to: CBS Library. Downloaded on March 31,2026 at 21:06:24 UTC from IEEE Xplore. Restrictions apply.